

HANIF EDMA FAUZ

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TECHNICAL SKILLS & LANGUAGES

- **Expertise:** Mobile Robotics, Control & Estimation, Electrical Engineering
- **Control & Estimation:** Model Predictive Control (MPC/MPPI/MPCC), PID, LQR, Kalman Filter (KF/EKF), 2D SLAM (mobile robot), Motion Planning & Trajectory Optimization
- **Machine Learning:** Gaussian Process (GP/SGP), Neural Networks, Imitation Learning, PyTorch & TensorFlow (basic)
- **Programming & Frameworks:** ROS1/ROS2, C/C++, Python, MATLAB/Simulink, CasADi, Flutter, HTML/CSS/JavaScript, OpenCV (basic)
- **Tools & Platforms:** Gazebo, move_base, Linux, Git, Docker
- **Hardware & IoT:** NVIDIA Jetson, Raspberry Pi, Arduino, ESP32
- **Spoken Languages:** Bahasa Indonesia (native), English (IELTS Academic Band 7.0, 2023), Korean (elementary)

EDUCATION

Seoul National University of Science and Technology (SEOULTECH) Seoul, South Korea
Master of Electrical and Information Engineering, 4.19/4.50 Feb 2024 – Aug 2026 (Expected)

- Recipient of SEOULTECH scholarship; Research Assistant; Special elective robotics and control engineering

Universitas Indonesia Depok, Indonesia
Bachelor of Electrical Engineering, 3.61/4.00 Aug 2019 – Aug 2023

- Special elective robotics and control engineering
- **Thesis:** “Design of Adaptive Extended Kalman Filter to Estimate State of Charge of Lithium-Ion Battery”

EXPERIENCE

Control & Intelligent Robotics Lab SEOULTECH (CIRL SEOULTECH) Seoul, South Korea
Research Assistant Feb 2024 – Present

- **[Published, 1st Author]** “Performance Enhanced Risk-Aware MPPI using Gaussian Process for Robust Mobile Robot Control”, *IEEE Access* (SCI Q1, 2025); integrated offline Gaussian Process (GP) uncertainty estimation with Risk-Aware MPPI (RA-MPPI), validated on FITENTH and Clearpath Jackal
- **[Submitted, 1st Author]** “Robust RA-MPPI using Online Learning”, submitted to IJCAS, Springer; online Sparse Gaussian Process (SGP) learning for real-time model correction and risk-aware trajectory sampling, applied to mobile robot control
- **[Ongoing]** Developing an imitation learning-based controller to replicate Model Predictive Contouring Control (MPCC) for mobile robot navigation, augmented with Disturbance Observer-Based (DOB) control
- **[Ongoing]** Developing a stability-guaranteed MPPI variant that samples feedback gain K instead of control input u , ensuring $u = -Kx$ satisfies local stability via closed-loop eigenvalue constraints

DoctorTool (PT Medifa Infoyasa Suryantara) Jakarta, Indonesia
IoT Engineer Intern Feb 2022 – Jul 2022

- Developed IoT medical devices to enhance measurement activities at posyandu, integrating various medical apparatus
- Engineered connectivity for over six categories of devices, such as thermometers, oximeters, and other biometric sensors

OTHER PROJECTS

6-DOF Robot Manipulator Simulation (OpenGL) Depok, Indonesia
Robotics Course Project – Universitas Indonesia May 2023

- Built a real-time 3D visualization of a 6-DOF robot arm in C/OpenGL (GLUT), implementing forward kinematics and Jacobian pseudoinverse-based inverse kinematics for end-effector position control
- Generated linear point-to-point trajectories and streamed joint commands to the physical robot over serial communication

Corrosion Detection Application Jakarta, Indonesia
1st Champion – SciHackfest 2023 (Hackathon), PT Sucofindo Sep 2023

- Developed a computer vision (CV) system for corrosion detection and built the Android version of the application using Flutter (compatible with Windows, Linux, and Android) within a 1-month timeframe; awarded 1st place and a prize of 25 million IDR (~1,600 USD) as a team of 3 people